

HABEEBAN MEMON

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[GitHub](#) | [LinkedIn](#) | [ResearchGate](#) | [Google Scholar](#)

RESEARCH PROFILE

Computer Systems Engineer with research experience in computer vision, machine learning, image processing, signal processing, and Non-Destructive Testing (NDT). Co-author of five research papers, including one published, one accepted, and three ready for submission, focused on defect detection, segmentation, thermographic imaging, depth estimation, 3D reconstruction, image super-resolution, multimodal learning and explainable ai for real-world engineering applications. Actively seeking thesis-based MSc research opportunities to collaborate with professors on applied AI and computer vision problems in intelligent inspection and visual systems, with a focus on building interpretable and trustworthy models for real-world engineering applications.

RESEARCH INTERESTS

Machine Learning and Deep Learning for Industrial Applications | Computer Vision and Image Processing | Non-Destructive Testing and Evaluation | Digital Signal Processing

EDUCATION

B.Eng. in Computer Systems Engineering (Result Awaited)

Sep 2022 - May 2026

Sukkur IBA University, Sukkur, Sindh, Pakistan | [iba-suk.edu.pk](#)

- CGPA: 3.5 / 4.00 (A = 4.0, US 4.0 GPA scale; all courses taught in English)
- PEC Accredited, Level II, Washington Accord Substantial Equivalence

PUBLICATIONS

Published

1. Mitigation of DDoS Attacks with Machine Learning, Deep Learning, and Transformers

2024

Zain Ul Abidin, Haseeb, **Habeeban Memon**, Muhammad Younis, Junaid Ahmed

International Journal of Multidisciplinary Conference Proceedings (IJMCP), Vol. 2(1), 2024 | DOI: [10.61503/Ijmcp.v2i1.181](#)

- Benchmarked ML and DL models for multi-class DDoS detection on CICDDoS2019. LightGBM and XGBoost achieved the best accuracy-speed balance for real-time network security.

Accepted (In Press)

2. Debond Detection in CFRP Structures using Optical Pulse Thermography and Deep Learning

2026

Habeeban Memon, Zain Ul Abidin, Junaid Ahmed, Ayesha Siddiqua

iCoMET 2025, IEEE Xplore | Preprint: [ResearchGate](#)

- Compared YOLOv8, Faster R-CNN, and Mask R-CNN for CFRP debond detection. Mask R-CNN achieved the highest accuracy with pixel-level defect segmentation.

Ready to Submit / Preprints

3. OPT-based Three-Channel Thermographic Fusion for Architecture-Independent Defect Detection and Segmentation in CFRP Composites

2026

Zain Ul Abidin, Habeeban Memon, Junaid Ahmed | Target journal to be confirmed

- Proposed an automated, architecture-independent input representation using a three-channel fusion based on cyclic alignment of raw thermal intensity, PCA, and TSR to isolate small, low-contrast defects.

4. Automated Depth Estimation and 3D Reconstruction of CFRP Subsurface Defects via Optical Pulsed Thermography

2026

Habeeban Memon, Zain Ul Abidin, Junaid Ahmed | Target journal to be confirmed

- Proposed a two-stage pipeline extracting 16 physics-informed features to predict through-thickness depth and reconstruct 3D topologies of CFRP subsurface delaminations.

5. CMM-DSE: Cross-Modal Consistency-Guided Multimodal Fake News Detection via Dempster-Shafer Evidential Reasoning

2026

Habeeban Memon, Zain Ul Abidin, Umair Ayaz Kamangar | Target journal to be confirmed

- Proposed a Cross-Modal Consistency Module and Dempster-Shafer evidential fusion layer, utilizing 2D FFT features for robust uncertainty quantification in multi-modal forgery detection.

ACADEMIC PROJECTS & THESIS

1. Undergraduate Thesis | Machine Learning Based Subsurface Defect Detection in CFRP Using Optical Pulse Thermography

May 2025 - May 2026

NDT/NDE | CFRP Composites | Defect Detection | Defect Segmentation | Depth Estimation | 3d Reconstruction

- Developed a machine learning-based system to automate the detection and analysis of subsurface defects in Carbon Fiber Reinforced Polymer (CFRP) composites.
- Processed raw optical pulsed thermography (OPT) sequences to build an end-to-end pipeline (Detection → Segmentation → Depth Estimation → 3D Reconstruction) for non-destructive testing.

- 2. Hospital Management System (CHMS)** 2026
ASP.NET | C# | SQL Server | JavaScript | Full-Stack | Live Deployment | [GitHub](#)
• Developed a web-based hospital management system for patient records, appointments, and billing, deployed live at chms.somee.com
- 3. Speech Enhancement through Advanced DSP Techniques** 2025
DSP | STFT/ISTFT | Spectral Subtraction | Noise Gating | [GitHub](#)
• Developed an end-to-end audio processing system using spectral subtraction and noise gating in the frequency domain to restore speech clarity in high-noise environments.
- 4. Image Compression Analysis: JPEG vs JPEG2000** 2024
Image Processing | DCT | DWT | PSNR | SSIM | [GitHub](#)
• Conducted performance analysis of DCT-based JPEG against wavelet-based JPEG2000, quantifying trade-offs between compression ratio and visual fidelity.
- 5. Smart Elevator Controller** 2025
FSM | Moore Machine | Manhattan Distance | Verilog | [GitHub](#)
• Developed a 5-floor digital elevator controller using Moore FSM with Manhattan distance-based nearest-request optimization.
- 6. IoT Health Monitor using ESP32 | 2025**
IoT | ESP32 | PulseSensor | LM35 | Blynk | [GitHub](#)
• Built a real-time health monitoring system with threshold-based emergency alerts and dual visualization on LCD and Blynk cloud.
- 7. Obstacle-Aware ROS2 Bot** 2025
ROS 2 | Gazebo | URDF | Sonar | Camera | [GitHub](#)
• Developed a mobile robot simulation with custom URDF robot and dual-sensor suite (sonar + camera) for obstacle-aware navigation.
- 8. MATLAB Voice Recognition using FFT** 2025
Signal Processing | FFT | MATLAB | Pitch Extraction | [GitHub](#)
• Developed a speaker identification system using FFT-based pitch extraction and distance-based matching for biometric security.
- 9. Student Management System with ML Integration** 2025
Desktop App | RBAC | MySQL | ML | Predictive Analytics | [GitHub](#)
• Engineered an academic administration desktop app with role-based access and ML-powered predictive analytics for student outcomes.
- 10. Real-Time Speech Enhancement System** 2025
DSP | Spectral Subtraction | Butterworth Filter | SNR Analysis | [GitHub](#)
• Implemented a multi-stage speech enhancement pipeline with noise spectrum estimation and Butterworth band-pass filtering for real-time noise reduction.

WORK EXPERIENCE

1. AI and Web Development Intern

Jul 2025 – Aug 2025

VitaNova – International Alliance for Sciences | Remote

- Integrated AI tools (ChatGPT, Perplexity, GravityWrite) into research and clinical diagnosis workflows, while applying Ethical AI frameworks to healthcare innovation and sustainable development goals.

2. Systems Engineering Intern

Jun 2025 – Jul 2025

State Life Insurance Corporation of Pakistan | On-Site, Sukkur, Sindh

- Conducted SWOT analysis of corporate digital infrastructure and analyzed departmental workflows to propose digitization strategies for manual insurance claim tracking.

TECHNICAL SKILLS & EXPERTISE

- **Programming & Development:** Python, MATLAB, SQL, Verilog, C#, HTML/CSS, ASP.NET Web Forms
- **Machine Learning & AI:** scikit-learn, XGBoost, LightGBM, Support Vector Regression (SVR), Random Forest
- **Deep Learning & Generative AI:** TensorFlow, PyTorch, Keras, ViT, DeBERTa-v3, BLIP, RT-DETR-L, UNet, FPN, DeepLabV3+, LSTM, LLMs, Prompt Engineering
- **Computer Vision:** OpenCV, YOLO, Faster R-CNN, Mask R-CNN, Semantic Segmentation, Instance Segmentation
- **Explainable AI (XAI):** SHAP, Grad-CAM, LIME
- **Scientific Computing & Data Analysis:** NumPy, SciPy, Pandas, Matplotlib, Seaborn
- **Web, Automation & Software Tools:** Streamlit, n8n, Git, GitHub, Jupyter Notebook
- **Databases:** SQL Server, MySQL, SQLite, ADO.NET
- **Embedded Systems & IoT:** ESP32, Arduino, Raspberry Pi (Basic)

CERTIFICATIONS

- Deep Learning Specialization: Advanced AI, Hands-on Lab, [Udemy](#), Udemy
 - Machine Learning Specialization, [DeepLearning.AI](#) / Coursera
 - Advanced Learning Algorithms, [DeepLearning.AI](#) / Coursera
 - Unsupervised Learning, Recommenders, Reinforcement Learning, [DeepLearning.AI](#) / Coursera
 - Supervised Machine Learning: Regression and Classification, [DeepLearning.AI](#) / Coursera
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